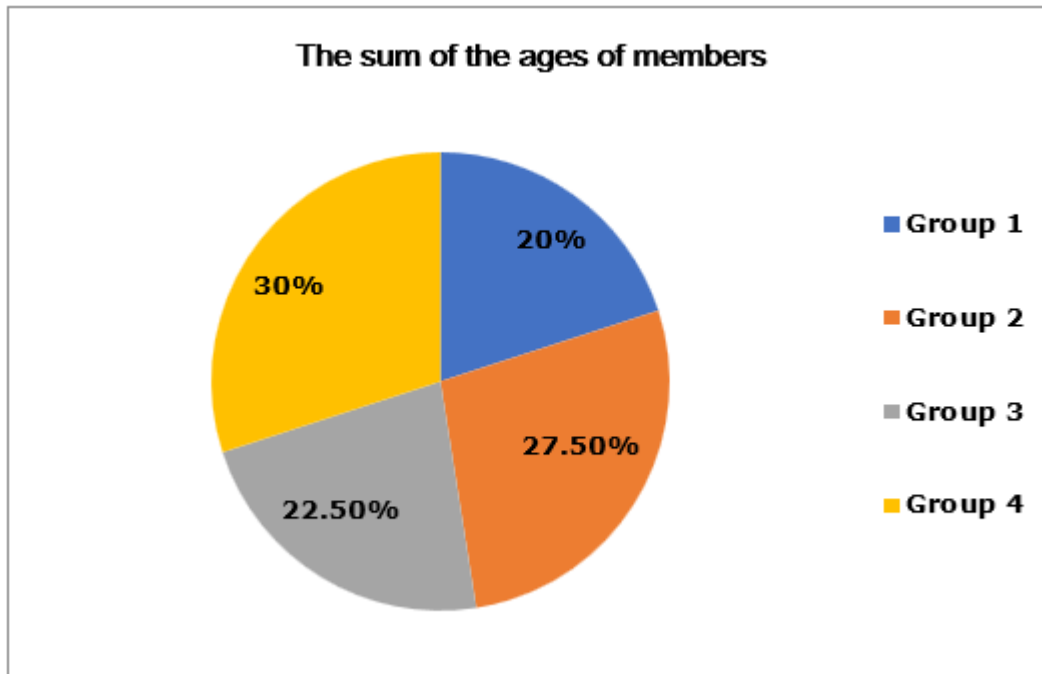


1. Questions

Study the following information carefully and answer the questions given below:

The pie chart given below shows the percentage distribution of the sum of the ages of members in four different groups out of the total ages of the four groups together.



Notes:

- i). The total age of the four groups together is 400 years.
- ii). The group members in each group are among A, B, C, D, E, and F.
- iii). Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.
- iv). Person C is not a member of group 3 only, and person D is not a member of group 1 only. Persons A, E, and B are members of two groups only.
- v). Persons B and E are not together in any group. Persons E and F are members of group 4, and persons A and E are members of group 3.

If the ratio of the age of B to the age of C is 5:7, and the age of A is 5 years younger than that of B, find the age of F.

- a. 20 years
- b. 25 years
- c. 15 years
- d. 10 years
- e. 30 years

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2. Questions

If the average age of persons A and E together is 20 years, find the difference between the ages of A

and D.

- a. 30 years
- b. 20 years
- c. 25 years
- d. 15 years
- e. 10 years

3. Questions

If group 5 has five members and the sum of their ages is 140 years, the members of group 5 are A and the members of group 4. Find the sum of the ages of persons C and F.

- a. 60 years
- b. 70 years
- c. 45 years
- d. 55 years
- e. 50 years

4. Questions

If the age of C is equal to the sum of the ages of E and F and the age of C is 35 years. Find the ratio of the age of A to that of B.

- a. 3: 4
- b. 1: 2
- c. 3: 2
- d. 4: 5
- e. 6: 5

5. Questions

The difference between the total ages of the members in group 1 and that of group 2 is what percentage of the total ages of the members in group 4?

- a. 20%
- b. 25%
- c. 30%
- d. 15%
- e. 10%

6. Questions

Study the following information carefully and answer the questions given below:

The table chart given below shows the number of devotees going to the temple by walking, the ratio of the number of devotees going to the temple by their own vehicle to that of government vehicles, and the total number of devotees in four different temples.

Temple	The number of devotees going to the temple by walking	The ratio of the number of devotees going to the temple by their own vehicle to that of government vehicles	The total number of devotees
A	$Y + 102$	17: 21	$X - 200$
B	$Y + 440$	19: 15	$X + 500$
C	$Y - 100$	7: 11	$X - 400$
D	$(X/4) + 751$	18: 15	$5Y + 150$

Notes:

- i). The total number of devotees = The number of devotees going to the temple (by walking + by their own vehicle + by government vehicles)
- ii). The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.
- iii). The number of devotees going to temple B by walking is 32% of the total number of devotees going to temple B.

If the ratio of the number of male to female devotees going to temple A by walking is 12:7, respectively. The number of male devotees going to temple A by their own vehicle and the number of male devotees going to temple A by government vehicle is 33.33% and 25% of the number of male devotees going to temple A by walking, respectively. Find the total number of female devotees going to temple A.

- a. 2698
- b. 1102
- c. 1207
- d. 1491
- e. 2234

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7. Questions

The difference between the number of devotees going to temples B and C by government vehicles is what percentage of the number of devotees going to temple C by walking?

- a. 66.66%

- b. 33.33%
- c. 11.11%
- d. 18.18%
- e. 44.44%

8. Questions

If the entry fee per devotee going to temple C by own vehicle and the ticket rate per devotee going to temple C by government vehicle are Rs. ____ and Rs. ____, respectively. If the total revenue generated from the devotees as mentioned above is Rs. 171000, then which of the following answers fill the blanks?

I: 80 and 54

II: 100 and 40

III: 60 and 65

- a. Only I
- b. Only III
- c. Both I and II
- d. Both II and III
- e. Only II

9. Questions

If the total number of devotees going to temple E is equal to the average number of devotees going to temples C and D together. The ratio of the number of devotees going to temple E by walking, by own vehicles, and by government vehicles is 5: 11: 9. Find the average number of devotees going to temples A, B, and E by walking.

- a. $(X/5) + 139$
- b. $Y - 139$
- c. $(X/5) + 129$
- d. $Y + 139$
- e. $X - Y - 1211$

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10. Questions

What is the difference between the total number of devotees going to temples by government vehicles together and the total number of devotees going to temples by their own vehicles together?

- a. 215
- b. 555

- c. 335
- d. 275
- e. 415

11. Questions

Study the following information carefully and answer the questions given below:

A and B started the business with some amount; after 'n' months, D joined with 90% of the initial investment of A, and A withdrew his investment by 20%. After 'm' more months, B increased his investment by 60%, C joined with 20% more than the initial investment of B, and D increased his investment by 25%. At the end of the two years, the profit ratios B, C, and D are 65: 30: 51, respectively.

Note:

- i). The value of 'n – m' is 8.

If the profit ratio of A and C is 106: 45 respectively, then find the initial investment amount of A.

- a. Rs. 30000
- b. Rs. 25000
- c. Rs. 40000
- d. Rs. 20000
- e. Cannot be determined

12. Questions

If the value of 'y' is 250 and A withdraws the amount from the business is Rs. P, which is invested at 15% compounded annually in the scheme. Find the interest received from the scheme after 3 years.

- a. Rs. 2580
- b. Rs. 1200
- c. Rs. 4167
- d. Rs. 5422
- e. Cannot be determined

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13. Questions

If the difference between the initial investment of C and that of D is Rs. 6000. Find the difference between the initial investment of C and that of A.

- a. Rs. 20000
- b. Rs. 12000
- c. Rs. 15000

- d. Rs. 5000
- e. Rs. 10000

14. Questions

If the total profit received in the business is Rs. 26000 and the profit share of A is Rs. 8480. Find the profit share of D.

- a. Rs. 4580
- b. Rs. 6120
- c. Rs. 66406
- d. Rs. 5420
- e. Cannot be determined

15. Questions

If D increased his investment amount in the business is Rs. 9000, then find that B increased his investment amount in the business.

- a. Rs. 10000
- b. Rs. 12000
- c. Rs. 15000
- d. Rs. 5000
- e. Rs. 20000

16. Questions

I: $x^2 - 5x - 84 = 0$

II: $y^2 - 25y + 136 = 0$

Which of the following equations is formed from the roots of the above-mentioned equations I, and II? (one from each equation)

- a. $z^2 - 10z + 119$
- b. $z^2 + 10z - 119$
- c. $z^2 + 4z - 96$
- d. $z^2 - 5z - 204$
- e. $z^2 - 29z + 204$

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17. Questions

$$3x^2 - 56x + L = 0$$

The roots of the given equation are M and N. The value of x is 14/3. Which of the following statements is true?

I: The HCF of L and M is 50% more than the value of N.

II: The value of L is a perfect square.

III: $L = M^2$

IV: $M > N$

- a. Only I
- b. Both I and III
- c. All I, II, and III
- d. Both II and III
- e. Only III

18. Questions

$$G + 2x^2 - (x - 7)^2 = 37x + 7$$

The smaller root of the equation is 9. Find the sum of G and the larger root of the equation.

- a. 196
- b. 172
- c. 182
- d. 168
- e. 191

19. Questions

Find the positive root of equation II, substitute it into equation I, and find the value of 'x'.

I: $(220.5/x) + (272.5/y) = 23.15$

II: $1 - (6/y) - (475/y^2) = 0$

- a. 24
- b. 12
- c. 16
- d. 18
- e. 20

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20. Questions

The highest root in equation I is the same as one of the roots of equation II. Find another root of equation II.

I: $x^2 - 7x - 330 = 0$

II: $y^2 - my + 462 = 0$

- a. 21
- b. 22
- c. 19
- d. 16
- e. 23

Explanations:

1. Questions

The total age of the four groups together = 400 years.

From Pie chart,

The sum of the ages of members in group 1 = 20% of 400 = 80 years

The sum of the ages of members in group 2 = 27.5% of 400 = 110 years

The sum of the ages of members in group 3 = 22.5% of 400 = 90 years

The sum of the ages of members in group 4 = 30% of 400 = 120 years

Let,

Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.

The total members in group 1 = 3

The total members in group 2 = 3

The total members in group 3 = 3

The total members in group 4 = 4

Let,

Person C is not a member of group 3 only, So, Person C is a member of groups 1, 2, and 4.

Person D is not a member of group 1 only, So, Person D is a member of groups 2, 3, and 4.

Persons E and F are members of group 4, and persons A and E are members of group 3.

So, the members of the groups

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Group	Members
Group 1	C
Group 2	C, D
Group 3	D, A, E
Group 4	C, D, E, F

Persons B and E are not together in any group. Persons A, E, and B are members of two groups only.

Group	Members
Group 1	C, B, A
Group 2	C, D, B
Group 3	D, A, E
Group 4	C, D, E, F

So,

The sum of age of group 1 = $A + B + C = 80$ years --> (1)

The sum of age of group 2 = $D + B + C = 110$ years --> (2)

The sum of age of group 3 = $A + D + E = 90$ years--> (3)

The sum of age of group 4 = $C + D + E + F = 120$ years --> (4)

Answer: C

According to the question,

The ratio of the age of B to the age of C = $5:7 = 5x: 7x$

The age of A = $5x - 5$ years

From (1)

$$5x - 5 + 5x + 7x = 80$$

$$17x = 85$$

$$x = 5$$

The age of A = $25 - 5 = 20$ years

The age of B = $5 * 5 = 25$ years

The age of C = $7 * 5 = 35$ years

From (2)

$$B + C + D = 110$$

$$25 + 35 + D = 110$$

$$D = 50$$

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The age of D = 50 years

From (3)

$$A + D + E = 90$$

$$20 + 50 + E = 90$$

$$E = 20$$

The age of E = 20 years

From (D)

$$C + D + E + F = 120$$

$$35 + 50 + 20 + F = 120$$

$$F = 15$$

The age of F = 15 years

Hence, the correct answer is option C.

2. Questions

The total age of the four groups together = 400 years.

From Pie chart,

The sum of the ages of members in group 1 = 20% of 400 = 80 years

The sum of the ages of members in group 2 = 27.5% of 400 = 110 years

The sum of the ages of members in group 3 = 22.5% of 400 = 90 years

The sum of the ages of members in group 4 = 30% of 400 = 120 years

Let,

Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.

The total members in group 1 = 3

The total members in group 2 = 3

The total members in group 3 = 3

The total members in group 4 = 4

Let,

Person C is not a member of group 3 only, So, Person C is a member of groups 1, 2, and 4.

Person D is not a member of group 1 only, So, Person D is a member of groups 2, 3, and 4.

Persons E and F are members of group 4, and persons A and E are members of group 3.

So, the members of the groups

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Group	Members
Group 1	C
Group 2	C, D
Group 3	D, A, E
Group 4	C, D, E, F

Persons B and E are not together in any group. Persons A, E, and B are members of two groups only.

Group	Members
Group 1	C, B, A
Group 2	C, D, B
Group 3	D, A, E
Group 4	C, D, E, F

So,

The sum of age of group 1 = $A + B + C = 80$ years --> (1)

The sum of age of group 2 = $D + B + C = 110$ years --> (2)

The sum of age of group 3 = $A + D + E = 90$ years --> (3)

The sum of age of group 4 = $C + D + E + F = 120$ years --> (4)

Answer: A

The total age of persons A and E together = $2 * 20 = 40$ years

From (3)

$$A + D + E = 90$$

$$D + 40 = 90$$

$$D = 50$$

The age of D = 50 years

From (2)

$$B + C + D = 110$$

$$B + C = 110 - 50$$

$$B + C = 60$$

From (1)

$$A + B + C = 80$$

$$A + 60 = 80$$

$$A = 20$$

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The age of A = 20 years

Thus, the required difference = $50 - 20 = 30$ years

Hence, the correct answer is option A.

3. Questions

The total age of the four groups together = 400 years.

From Pie chart,

The sum of the ages of members in group 1 = 20% of 400 = 80 years

The sum of the ages of members in group 2 = 27.5% of 400 = 110 years

The sum of the ages of members in group 3 = 22.5% of 400 = 90 years

The sum of the ages of members in group 4 = 30% of 400 = 120 years

Let,

Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.

The total members in group 1 = 3

The total members in group 2 = 3

The total members in group 3 = 3

The total members in group 4 = 4

Let,

Person C is not a member of group 3 only, So, Person C is a member of groups 1, 2, and 4.

Person D is not a member of group 1 only, So, Person D is a member of groups 2, 3, and 4.

Persons E and F are members of group 4, and persons A and E are members of group 3.

So, the members of the groups

Group	Members
Group 1	C
Group 2	C, D
Group 3	D, A, E
Group 4	C, D, E, F

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Persons B and E are not together in any group. Persons A, E, and B are members of two groups only.

Group	Members
Group 1	C, B, A
Group 2	C, D, B
Group 3	D, A, E
Group 4	C, D, E, F

So,

The sum of age of group 1 = $A + B + C = 80$ years --> (1)

The sum of age of group 2 = $D + B + C = 110$ years --> (2)

The sum of age of group 3 = $A + D + E = 90$ years --> (3)

The sum of age of group 4 = $C + D + E + F = 120$ years --> (4)

Answer: E

The members of group 5 = A and the members of group 4 = A, C, D, E, F

So,

$$A + C + D + E + F = 140$$

$$A + 120 = 140$$

The age of A = 20 years

From (3)

$$A + D + E = 90$$

$$20 + D + E = 90$$

$$D + E = 70$$

From (4)

$$C + D + E + F = 120$$

$$C + F + 70 = 120$$

$$C + F = 50$$

The sum of the ages of persons C and F = 50 years

Hence, the correct answer is option E.

4. Questions

The total age of the four groups together = 400 years.

From Pie chart,

The sum of the ages of members in group 1 = 20% of 400 = 80 years

The sum of the ages of members in group 2 = 27.5% of 400 = 110 years

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The sum of the ages of members in group 3 = 22.5% of 400 = 90 years

The sum of the ages of members in group 4 = 30% of 400 = 120 years

Let,

Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.

The total members in group 1 = 3

The total members in group 2 = 3

The total members in group 3 = 3

The total members in group 4 = 4

Let,

Person C is not a member of group 3 only, So, Person C is a member of groups 1, 2, and 4.

Person D is not a member of group 1 only, So, Person D is a member of groups 2, 3, and 4.

Persons E and F are members of group 4, and persons A and E are members of group 3.

So, the members of the groups

Group	Members
Group 1	C
Group 2	C, D
Group 3	D, A, E
Group 4	C, D, E, F

Persons B and E are not together in any group. Persons A, E, and B are members of two groups only.

Group	Members
Group 1	C, B, A
Group 2	C, D, B
Group 3	D, A, E
Group 4	C, D, E, F

So,

The sum of age of group 1 = $A + B + C = 80$ years --> (1)

The sum of age of group 2 = $D + B + C = 110$ years --> (2)

The sum of age of group 3 = $A + D + E = 90$ years --> (3)

The sum of age of group 4 = $C + D + E + F = 120$ years --> (4)

Answer: D

$C + D + E + F = 120$ ---> (4)

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$$B + C + D = 110 \rightarrow (2)$$

Solve (2) and (4)

$$E + F - B = 10 \rightarrow (5)$$

The age of C is equal to the sum of the ages of E and F

$$C = E + F \rightarrow (6)$$

Equation (6) is sub into (5)

$$C - B = 10$$

The age of C = 35 years

$$B = 35 - 10$$

The age of B = 25 years

From (1)

$$A + 25 + 35 = 80$$

$$A = 20$$

The age of A = 20 years

Thus, the required ratio = 20: 25 = 4: 5

Hence, the correct answer is option D.

5. Questions

The total age of the four groups together = 400 years.

From Pie chart,

The sum of the ages of members in group 1 = 20% of 400 = 80 years

The sum of the ages of members in group 2 = 27.5% of 400 = 110 years

The sum of the ages of members in group 3 = 22.5% of 400 = 90 years

The sum of the ages of members in group 4 = 30% of 400 = 120 years

Let,

Groups 1, 2, and 3 have only 3 members, while the rest of the groups have 4 members.

The total members in group 1 = 3

The total members in group 2 = 3

The total members in group 3 = 3

The total members in group 4 = 4

Let,

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Person C is not a member of group 3 only, So, Person C is a member of groups 1, 2, and 4.

Person D is not a member of group 1 only, So, Person D is a member of groups 2, 3, and 4.

Persons E and F are members of group 4, and persons A and E are members of group 3.

So, the members of the groups

Group	Members
Group 1	C
Group 2	C, D
Group 3	D, A, E
Group 4	C, D, E, F

Persons B and E are not together in any group. Persons A, E, and B are members of two groups only.

Group	Members
Group 1	C, B, A
Group 2	C, D, B
Group 3	D, A, E
Group 4	C, D, E, F

So,

The sum of age of group 1 = $A + B + C = 80$ years --> (1)

The sum of age of group 2 = $D + B + C = 110$ years --> (2)

The sum of age of group 3 = $A + D + E = 90$ years --> (3)

The sum of age of group 4 = $C + D + E + F = 120$ years --> (4)

Answer: B

The difference between the total ages of the members in group 1 and that of group 2 = $110 - 80 = 30$ years

The sum of the ages of group 4 = 120 years

Thus, the required percentage = $(30/120) * 100 = 25\%$

Hence, the correct answer is option B.

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6. Questions

The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.

$$(X - 200) * 29/100 = Y + 102$$

$$29X - 5800 = 100Y + 10200$$

$$29X - 100Y = 16000 \text{ ---> (1)}$$

The number of devotees going to temple B by walking is 32% of the total number of devotees going to temple B.

$$(X + 500) * 32/100 = Y + 440$$

$$32X + 16000 = 100Y + 44000$$

$$32X - 100Y = 28000 \text{ ---> (2)}$$

$$(2) - (1)$$

$$32X - 29X = 28000 - 16000$$

$$3X = 12000$$

$$X = 4000$$

And

From (1)

$$29 * 4000 - 100Y = 16000$$

$$100Y = 100000$$

$$Y = 1000$$

In temple A

$$\text{The total number of devotees} = X - 200 = 4000 - 200 = 3800$$

$$\text{The number of devotees going to the temple by walking} = Y + 102 = 1102$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3800 - 1102 = 2698$$

$$\text{The number of devotees going to the temple by their own vehicle} = (2698/38) * 17 = 1207$$

$$\text{The number of devotees going to the temple by government vehicles} = (2698/38) * 21 = 1491$$

In temple B

$$\text{The total number of devotees} = X + 500 = 4000 + 500 = 4500$$

$$\text{The number of devotees going to the temple by walking} = Y + 440 = 1440$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 4500 - 1440 = 3060$$

$$\text{The number of devotees going to the temple by their own vehicle} = (3060/34) * 19 = 1710$$

$$\text{The number of devotees going to the temple by government vehicles} = (3060/34) * 15 = 1350$$

In temple C

$$\text{The total number of devotees} = X - 400 = 4000 - 400 = 3600$$

$$\text{The number of devotees going to the temple by walking} = Y - 100 = 900$$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $3600 - 900 = 2700$

The number of devotees going to the temple by their own vehicle = $(2700/18) * 7 = 1050$

The number of devotees going to the temple by government vehicles = $(2700/18) * 11 = 1650$

In temple D

The total number of devotees = $5Y + 150 = 5000 + 150 = 5150$

The number of devotees going to the temple by walking = $(X/4) + 751 = 1000 + 751 = 1751$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $5150 - 1751 = 3399$

The number of devotees going to the temple by their own vehicle = $(3399/33) * 18 = 1854$

The number of devotees going to the temple by government vehicles = $(3399/33) * 15 = 1545$

Let table

Temple	The number of devotees going to the temple by walking	The number of devotees going to the temple by their own vehicle	The number of devotees going to the temple by government vehicles	The total number of devotees
A	1102	1207	1491	3800
B	1440	1710	1350	4500
C	900	1050	1650	3600
D	1751	1854	1545	5150

Answer: A

According to the question,

The number of males devotees going to temple A by walking = $(1102/19) * 12 = 696$

The number of male devotees going to temple A by their own vehicle = 33.33% of $696 = (1/3) * 696 = 232$

The number of male devotees going to temple A by government vehicles = 25% of $696 = 174$

The total number of male devotees going to temple A = $696 + 232 + 174 = 1102$

The total number of female devotees going to temple A = $3800 - 696 - 232 - 174 = 2698$

Hence, the correct answer is option A.

7. Questions

The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.

$$(X - 200) * 29/100 = Y + 102$$

$$29X - 5800 = 100Y + 10200$$

$$29X - 100Y = 16000 \text{ ---> (1)}$$

The number of devotees going to temple B by walking is 32% of the total number of devotees going to temple B.

$$(X + 500) * 32/100 = Y + 440$$

$$32X + 16000 = 100Y + 44000$$

$$32X - 100Y = 28000 \text{ ---> (2)}$$

$$(2) - (1)$$

$$32X - 29X = 28000 - 16000$$

$$3X = 12000$$

$$X = 4000$$

And

From (1)

$$29 * 4000 - 100Y = 16000$$

$$100Y = 100000$$

$$Y = 1000$$

In temple A

$$\text{The total number of devotees} = X - 200 = 4000 - 200 = 3800$$

$$\text{The number of devotees going to the temple by walking} = Y + 102 = 1102$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3800 - 1102 = 2698$$

$$\text{The number of devotees going to the temple by their own vehicle} = (2698/38) * 17 = 1207$$

$$\text{The number of devotees going to the temple by government vehicles} = (2698/38) * 21 = 1491$$

In temple B

$$\text{The total number of devotees} = X + 500 = 4000 + 500 = 4500$$

$$\text{The number of devotees going to the temple by walking} = Y + 440 = 1440$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 4500 - 1440 = 3060$$

$$\text{The number of devotees going to the temple by their own vehicle} = (3060/34) * 19 = 1710$$

$$\text{The number of devotees going to the temple by government vehicles} = (3060/34) * 15 = 1350$$

In temple C

$$\text{The total number of devotees} = X - 400 = 4000 - 400 = 3600$$

The number of devotees going to the temple by walking = $Y - 100 = 900$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $3600 - 900 = 2700$

The number of devotees going to the temple by their own vehicle = $(2700/18) * 7 = 1050$

The number of devotees going to the temple by government vehicles = $(2700/18) * 11 = 1650$

In temple D

The total number of devotees = $5Y + 150 = 5000 + 150 = 5150$

The number of devotees going to the temple by walking = $(X/4) + 751 = 1000 + 751 = 1751$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $5150 - 1751 = 3399$

The number of devotees going to the temple by their own vehicle = $(3399/33) * 18 = 1854$

The number of devotees going to the temple by government vehicles = $(3399/33) * 15 = 1545$

Let table

Temple	The number of devotees going to the temple by walking	The number of devotees going to the temple by their own vehicle	The number of devotees going to the temple by government vehicles	The total number of devotees
A	1102	1207	1491	3800
B	1440	1710	1350	4500
C	900	1050	1650	3600
D	1751	1854	1545	5150

Answer: B

The difference between the number of devotees going to temples B and C by government vehicles = $1650 - 1350 = 300$

Thus, required percentage = $(300/900) * 100 = 33.33\%$

Hence, the correct answer is option B.

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8. Questions

The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.

$$(X - 200) * 29/100 = Y + 102$$

$$29X - 5800 = 100Y + 10200$$

$$29X - 100Y = 16000 \text{ ---> (1)}$$

The number of devotees going to temple B by walking is 32% of the total number of devotees going to

temple B.

$$(X + 500) * 32/100 = Y + 440$$

$$32X + 16000 = 100Y + 44000$$

$$32X - 100Y = 28000 \text{ ---} \rightarrow (2)$$

$$(2) - (1)$$

$$32X - 29X = 28000 - 16000$$

$$3X = 12000$$

$$X = 4000$$

And

From (1)

$$29 * 4000 - 100Y = 16000$$

$$100Y = 100000$$

$$Y = 1000$$

In temple A

$$\text{The total number of devotees} = X - 200 = 4000 - 200 = 3800$$

$$\text{The number of devotees going to the temple by walking} = Y + 102 = 1102$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3800 - 1102 = 2698$$

$$\text{The number of devotees going to the temple by their own vehicle} = (2698/38) * 17 = 1207$$

$$\text{The number of devotees going to the temple by government vehicles} = (2698/38) * 21 = 1491$$

In temple B

$$\text{The total number of devotees} = X + 500 = 4000 + 500 = 4500$$

$$\text{The number of devotees going to the temple by walking} = Y + 440 = 1440$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 4500 - 1440 = 3060$$

$$\text{The number of devotees going to the temple by their own vehicle} = (3060/34) * 19 = 1710$$

$$\text{The number of devotees going to the temple by government vehicles} = (3060/34) * 15 = 1350$$

In temple C

$$\text{The total number of devotees} = X - 400 = 4000 - 400 = 3600$$

$$\text{The number of devotees going to the temple by walking} = Y - 100 = 900$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3600 - 900 = 2700$$

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The number of devotees going to the temple by their own vehicle = $(2700/18) * 7 = 1050$

The number of devotees going to the temple by government vehicles = $(2700/18) * 11 = 1650$

In temple D

The total number of devotees = $5Y + 150 = 5000 + 150 = 5150$

The number of devotees going to the temple by walking = $(X/4) + 751 = 1000 + 751 = 1751$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $5150 - 1751 = 3399$

The number of devotees going to the temple by their own vehicle = $(3399/33) * 18 = 1854$

The number of devotees going to the temple by government vehicles = $(3399/33) * 15 = 1545$

Let table

Temple	The number of devotees going to the temple by walking	The number of devotees going to the temple by their own vehicle	The number of devotees going to the temple by government vehicles	The total number of devotees
A	1102	1207	1491	3800
B	1440	1710	1350	4500
C	900	1050	1650	3600
D	1751	1854	1545	5150

Answer: E

I: 80 and 54

Let

The total revenue generated = $1050 * 80 + 1650 * 54 = \text{Rs. } 173100$

$173100 \neq 171000$

This is not satisfied.

II: 100 and 40

Let

The total revenue generated = $1050 * 100 + 1650 * 40 = \text{Rs. } 171000$

$171000 = 171000$

This is satisfied.

III: 60 and 65

Let

The total revenue generated = $1050 * 60 + 1650 * 65 = \text{Rs. } 170250$

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170250 $\neq$ 171000

This is not satisfied.

Hence, the correct answer is option E.

9. Questions

The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.

$$(X - 200) * 29/100 = Y + 102$$

$$29X - 5800 = 100Y + 10200$$

$$29X - 100Y = 16000 \text{ ---> (1)}$$

The number of devotees going to temple B by walking is 32% of the total number of devotees going to temple B.

$$(X + 500) * 32/100 = Y + 440$$

$$32X + 16000 = 100Y + 44000$$

$$32X - 100Y = 28000 \text{ ---> (2)}$$

$$(2) - (1)$$

$$32X - 29X = 28000 - 16000$$

$$3X = 12000$$

$$X = 4000$$

And

From (1)

$$29 * 4000 - 100Y = 16000$$

$$100Y = 100000$$

$$Y = 1000$$

In temple A

$$\text{The total number of devotees} = X - 200 = 4000 - 200 = 3800$$

$$\text{The number of devotees going to the temple by walking} = Y + 102 = 1102$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3800 - 1102 = 2698$$

$$\text{The number of devotees going to the temple by their own vehicle} = (2698/38) * 17 = 1207$$

$$\text{The number of devotees going to the temple by government vehicles} = (2698/38) * 21 = 1491$$

In temple B

$$\text{The total number of devotees} = X + 500 = 4000 + 500 = 4500$$

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The number of devotees going to the temple by walking = $Y + 440 = 1440$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $4500 - 1440 = 3060$

The number of devotees going to the temple by their own vehicle = $(3060/34) * 19 = 1710$

The number of devotees going to the temple by government vehicles = $(3060/34) * 15 = 1350$

In temple C

The total number of devotees = $X - 400 = 4000 - 400 = 3600$

The number of devotees going to the temple by walking = $Y - 100 = 900$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $3600 - 900 = 2700$

The number of devotees going to the temple by their own vehicle = $(2700/18) * 7 = 1050$

The number of devotees going to the temple by government vehicles = $(2700/18) * 11 = 1650$

In temple D

The total number of devotees = $5Y + 150 = 5000 + 150 = 5150$

The number of devotees going to the temple by walking = $(X/4) + 751 = 1000 + 751 = 1751$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $5150 - 1751 = 3399$

The number of devotees going to the temple by their own vehicle = $(3399/33) * 18 = 1854$

The number of devotees going to the temple by government vehicles = $(3399/33) * 15 = 1545$

Let table

Temple	The number of devotees going to the temple by walking	The number of devotees going to the temple by their own vehicle	The number of devotees going to the temple by government vehicles	The total number of devotees
A	1102	1207	1491	3800
B	1440	1710	1350	4500
C	900	1050	1650	3600
D	1751	1854	1545	5150

Answer: D

The total number of devotees going to temple E = $(3600 + 5150)/2 = 4375$

The number of devotees going to temple E by walking = $(4375/25) * 5 = 875$

Thus, required average = $(1102 + 1440 + 875)/3 = 3417/3 = 1139$

$Y + 139 = 1000 + 139 = 1139$

Hence, the correct answer is option D.

10. Questions

The number of devotees going to temple A by walking is 29% of the total number of devotees going to temple A.

$$(X - 200) * 29/100 = Y + 102$$

$$29X - 5800 = 100Y + 10200$$

$$29X - 100Y = 16000 \text{ ---> (1)}$$

The number of devotees going to temple B by walking is 32% of the total number of devotees going to temple B.

$$(X + 500) * 32/100 = Y + 440$$

$$32X + 16000 = 100Y + 44000$$

$$32X - 100Y = 28000 \text{ ---> (2)}$$

$$(2) - (1)$$

$$32X - 29X = 28000 - 16000$$

$$3X = 12000$$

$$X = 4000$$

And

From (1)

$$29 * 4000 - 100Y = 16000$$

$$100Y = 100000$$

$$Y = 1000$$

In temple A

$$\text{The total number of devotees} = X - 200 = 4000 - 200 = 3800$$

$$\text{The number of devotees going to the temple by walking} = Y + 102 = 1102$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 3800 - 1102 = 2698$$

$$\text{The number of devotees going to the temple by their own vehicle} = (2698/38) * 17 = 1207$$

$$\text{The number of devotees going to the temple by government vehicles} = (2698/38) * 21 = 1491$$

In temple B

$$\text{The total number of devotees} = X + 500 = 4000 + 500 = 4500$$

$$\text{The number of devotees going to the temple by walking} = Y + 440 = 1440$$

$$\text{The number of devotees going to the temple (by their own vehicle + by government vehicles)} = 4500 -$$

$$1440 = 3060$$

The number of devotees going to the temple by their own vehicle = $(3060/34) * 19 = 1710$

The number of devotees going to the temple by government vehicles = $(3060/34) * 15 = 1350$

In temple C

The total number of devotees = $X - 400 = 4000 - 400 = 3600$

The number of devotees going to the temple by walking = $Y - 100 = 900$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $3600 - 900 = 2700$

The number of devotees going to the temple by their own vehicle = $(2700/18) * 7 = 1050$

The number of devotees going to the temple by government vehicles = $(2700/18) * 11 = 1650$

In temple D

The total number of devotees = $5Y + 150 = 5000 + 150 = 5150$

The number of devotees going to the temple by walking = $(X/4) + 751 = 1000 + 751 = 1751$

The number of devotees going to the temple (by their own vehicle + by government vehicles) = $5150 - 1751 = 3399$

The number of devotees going to the temple by their own vehicle = $(3399/33) * 18 = 1854$

The number of devotees going to the temple by government vehicles = $(3399/33) * 15 = 1545$

Let table

Temple	The number of devotees going to the temple by walking	The number of devotees going to the temple by their own vehicle	The number of devotees going to the temple by government vehicles	The total number of devotees
A	1102	1207	1491	3800
B	1440	1710	1350	4500
C	900	1050	1650	3600
D	1751	1854	1545	5150

Answer: A

The total number of devotees going to temples by government vehicles together= $1491 + 1350 + 1650 + 1545 = 6036$

The total number of devotees going to temples by their own vehicles together= $1207 + 1710 + 1050 + 1854 = 5821$

Thus, the required difference = $6036 - 5821 = 215$

Hence, the correct answer is option A.

11. Questions

Let the initial investment of A and B be Rs. 100x, and Rs. 100y respectively.

The profit share of A

$$= 100x * n + 80\% \text{ of } 100x * (24 - n)$$

$$= 100xn + 1920x - 80xn$$

$$= 20xn + 1920x$$

The profit share of B

$$= 100y * (n + m) + 160\% \text{ of } 100y * (24 - n - m)$$

$$= 100yn + 100ym + 3840y - 160yn - 160ym$$

$$= 3840y - 60yn - 60ym$$

The profit share of C

$$= 120\% \text{ of } 100y * (24 - n - m)$$

$$= 2880y - 120yn - 120ym$$

The profit share of D

$$= 90\% \text{ of } 100x * m + 125\% \text{ of } (90\% \text{ of } 100x) * (24 - n - m)$$

$$= 90xm + 2700x - 112.5xn - 112.5xm$$

$$= 2700x - 112.5xn - 22.5xm$$

Let

The profit ratios B, C, and D are 65: 30: 51

We take that B and C

$$(3840y - 60yn - 60ym)/(2880y - 120yn - 120ym) = 65/30$$

$$23040 - 360n - 360m = 37440 - 1560n - 1560m$$

$$1200n + 1200m = 14400$$

$$n + m = 12 \text{ ---> (1)}$$

The value of 'n - m' is 8.

$$n - m = 8 \text{ ---> (2)}$$

Solve (1) and (2)

$$2n = 20$$

$$n = 10$$

and

$$m = 2$$

@QuizByMickey

We take that C and D

$$(2880y - 120yn - 120ym)/(2700x - 112.5xn - 22.5xm) = 30/51$$

$$(960y - 40y * 10 - 40y * 2)/(900x - 37.5x * 10 - 7.5x * 2) = 10/17$$

$$(48y)/(510x) = 1/17$$

$$48y = 30x$$

$$x/y = 48/30$$

$$x/y = 8/5$$

Answer: E

Let, WKT

According to the question,

$$(20xn + 1920x)/(2880y - 120yn - 120ym) = 106/45$$

Substitute 'n' and 'm' values

$$2120x/1440y = 106/45$$

$$2x/16y = 1/5$$

$$x/y = 8/5$$

We cannot find the exact values of x and y.

So, we cannot find the initial investment amount of A.

Hence, the correct answer is option E.

12. Questions

Let the initial investment of A and B be Rs. 100x, and Rs. 100y respectively.

The profit share of A

$$= 100x * n + 80\% \text{ of } 100x * (24 - n)$$

$$= 100xn + 1920x - 80xn$$

$$= 20xn + 1920x$$

The profit share of B

$$= 100y * (n + m) + 160\% \text{ of } 100y * (24 - n - m)$$

$$= 100yn + 100ym + 3840y - 160yn - 160ym$$

$$= 3840y - 60yn - 60ym$$

The profit share of C

$$= 120\% \text{ of } 100y * (24 - n - m)$$

$$= 2880y - 120yn - 120ym$$

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The profit share of D

$$= 90\% \text{ of } 100x * m + 125\% \text{ of } (90\% \text{ of } 100x) * (24 - n - m)$$

$$= 90xm + 2700x - 112.5xn - 112.5xm$$

$$= 2700x - 112.5xn - 22.5xm$$

Let

The profit ratios B, C, and D are 65: 30: 51

We take that B and C

$$(3840y - 60yn - 60ym)/(2880y - 120yn - 120ym) = 65/30$$

$$23040 - 360n - 360m = 37440 - 1560n - 1560m$$

$$1200n + 1200m = 14400$$

$$n + m = 12 \text{ ---> (1)}$$

The value of 'n - m' is 8.

$$n - m = 8 \text{ ---> (2)}$$

Solve (1) and (2)

$$2n = 20$$

$$n = 10$$

and

$$m = 2$$

We take that C and D

$$(2880y - 120yn - 120ym)/(2700x - 112.5xn - 22.5xm) = 30/51$$

$$(960y - 40y * 10 - 40y * 2)/(900x - 37.5x * 10 - 7.5x * 2) = 10/17$$

$$(48y)/(510x) = 1/17$$

$$48y = 30x$$

$$x/y = 48/30$$

$$x/y = 8/5$$

Answer: C

WKT

$$x/y = 8/5$$

$$x = (8/5) * 250$$

$$x = 400$$

So, the initial investment of A is = $100x$ = Rs. 40000

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A withdraw's amount from business = 20% of 40000 = Rs. 8000

Let,

$$\text{The interest received} = 8000 * (115/100)^3 - 8000$$

$$= 8000 * [(23/20)^3 - 1]$$

$$= 8000 * 4167/8000$$

$$= \text{Rs. } 4167$$

Hence, the correct answer is option C.

13. Questions

Let the initial investment of A and B be Rs. 100x, and Rs. 100y respectively.

The profit share of A

$$= 100x * n + 80\% \text{ of } 100x * (24 - n)$$

$$= 100xn + 1920x - 80xn$$

$$= 20xn + 1920x$$

The profit share of B

$$= 100y * (n + m) + 160\% \text{ of } 100y * (24 - n - m)$$

$$= 100yn + 100ym + 3840y - 160yn - 160ym$$

$$= 3840y - 60yn - 60ym$$

The profit share of C

$$= 120\% \text{ of } 100y * (24 - n - m)$$

$$= 2880y - 120yn - 120ym$$

The profit share of D

$$= 90\% \text{ of } 100x * m + 125\% \text{ of } (90\% \text{ of } 100x) * (24 - n - m)$$

$$= 90xm + 2700x - 112.5xn - 112.5xm$$

$$= 2700x - 112.5xn - 22.5xm$$

Let

The profit ratios B, C, and D are 65: 30: 51

We take that B and C

$$(3840y - 60yn - 60ym)/(2880y - 120yn - 120ym) = 65/30$$

$$23040 - 360n - 360m = 37440 - 1560n - 1560m$$

$$1200n + 1200m = 14400$$

$$n + m = 12 \text{ ---> (1)}$$

@QuizByMickey

The value of 'n – m' is 8.

$$n - m = 8 \text{ ---> (2)}$$

Solve (1) and (2)

$$2n = 20$$

$$n = 10$$

and

$$m = 2$$

We take that C and D

$$(2880y - 120yn - 120ym)/(2700x - 112.5xn - 22.5xm) = 30/51$$

$$(960y - 40y * 10 - 40y * 2)/(900x - 37.5x * 10 - 7.5x * 2) = 10/17$$

$$(48y)/(510x) = 1/17$$

$$48y = 30x$$

$$x/y = 48/30$$

$$x/y = 8/5$$

Answer: E

The initial investment of C = 120% of 100y = 120y

The initial investment of D = 90% of 100x = 90x

WKT

$$x/y = 8/5$$

$$x = 8y/5$$

Let

The initial investment of D = $90 * 8y/5 = 144y$

$$144y - 120y = 6000$$

$$24y = 6000$$

$$y = 250$$

$$x = 8 * 250/5 = 400$$

Then

The initial investment of C = $120 * 250 = \text{Rs. } 30000$

The initial investment of A = $100x = \text{Rs. } 40000$

Thus, the required difference = $40000 - 30000 = \text{Rs. } 10000$

Hence, the correct answer is option E.

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14. Questions

Let the initial investment of A and B be Rs. 100x, and Rs. 100y respectively.

The profit share of A

$$= 100x * n + 80\% \text{ of } 100x * (24 - n)$$

$$= 100xn + 1920x - 80xn$$

$$= 20xn + 1920x$$

The profit share of B

$$= 100y * (n + m) + 160\% \text{ of } 100y * (24 - n - m)$$

$$= 100yn + 100ym + 3840y - 160yn - 160ym$$

$$= 3840y - 60yn - 60ym$$

The profit share of C

$$= 120\% \text{ of } 100y * (24 - n - m)$$

$$= 2880y - 120yn - 120ym$$

The profit share of D

$$= 90\% \text{ of } 100x * m + 125\% \text{ of } (90\% \text{ of } 100x) * (24 - n - m)$$

$$= 90xm + 2700x - 112.5xn - 112.5xm$$

$$= 2700x - 112.5xn - 22.5xm$$

Let

The profit ratios B, C, and D are 65: 30: 51

We take that B and C

$$(3840y - 60yn - 60ym) / (2880y - 120yn - 120ym) = 65/30$$

$$23040 - 360n - 360m = 37440 - 1560n - 1560m$$

$$1200n + 1200m = 14400$$

$$n + m = 12 \text{ ---> (1)}$$

The value of 'n - m' is 8.

$$n - m = 8 \text{ ---> (2)}$$

Solve (1) and (2)

$$2n = 20$$

$$n = 10$$

and

$$m = 2$$

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We take that C and D

$$(2880y - 120yn - 120ym)/(2700x - 112.5xn - 22.5xm) = 30/51$$

$$(960y - 40y * 10 - 40y * 2)/(900x - 37.5x * 10 - 7.5x * 2) = 10/17$$

$$(48y)/(510x) = 1/17$$

$$48y = 30x$$

$$x/y = 48/30$$

$$x/y = 8/5$$

Answer: B

The sum of the profit share of B, C, and D together = 26000 – 8480 = Rs. 17520

The profit share of D = (17520/146) * 51 = Rs. 6120

Hence, the correct answer is option B.

15. Questions

Let the initial investment of A and B be Rs. 100x, and Rs. 100y respectively.

The profit share of A

$$= 100x * n + 80\% \text{ of } 100x * (24 - n)$$

$$= 100xn + 1920x - 80xn$$

$$= 20xn + 1920x$$

The profit share of B

$$= 100y * (n + m) + 160\% \text{ of } 100y * (24 - n - m)$$

$$= 100yn + 100ym + 3840y - 160yn - 160ym$$

$$= 3840y - 60yn - 60ym$$

The profit share of C

$$= 120\% \text{ of } 100y * (24 - n - m)$$

$$= 2880y - 120yn - 120ym$$

The profit share of D

$$= 90\% \text{ of } 100x * m + 125\% \text{ of } (90\% \text{ of } 100x) * (24 - n - m)$$

$$= 90xm + 2700x - 112.5xn - 112.5xm$$

$$= 2700x - 112.5xn - 22.5xm$$

Let

The profit ratios B, C, and D are 65: 30: 51

We take that B and C

@QuizByMickey

$$(3840y - 60yn - 60ym)/(2880y - 120yn - 120ym) = 65/30$$

$$23040 - 360n - 360m = 37440 - 1560n - 1560m$$

$$1200n + 1200m = 14400$$

$$n + m = 12 \text{ ---> (1)}$$

The value of 'n - m' is 8.

$$n - m = 8 \text{ ---> (2)}$$

Solve (1) and (2)

$$2n = 20$$

$$n = 10$$

and

$$m = 2$$

We take that C and D

$$(2880y - 120yn - 120ym)/(2700x - 112.5xn - 22.5xm) = 30/51$$

$$(960y - 40y * 10 - 40y * 2)/(900x - 37.5x * 10 - 7.5x * 2) = 10/17$$

$$(48y)/(510x) = 1/17$$

$$48y = 30x$$

$$x/y = 48/30$$

$$x/y = 8/5$$

Answer: C

D increased his investment amount in the business = 125% of (90% of 100x) - 90% of 100x = 112.5x - 90x = 22.5x

$$22.5x = 9000$$

$$x = 400$$

and

$$y = (400) * (5/8)$$

$$y = 250$$

B increased his investment amount in the business = 160% of 100y - 100y = 60y = 60 * 250 = Rs. 15000

Hence, the correct answer is option C.

16. Questions

Answer: E

$$\text{I: } x^2 - 5x - 84 = 0$$

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Let

$$x^2 - 12x + 7x - 84 = 0$$

$$x(x - 12) + 7(x - 12) = 0$$

$$(x - 12)(x + 7) = 0$$

$$x = +12 \text{ and } -7$$

$$\text{II: } y^2 - 25y + 136 = 0$$

Let

$$y^2 - 17y - 8y + 136 = 0$$

$$y(y - 17) - 8(y - 17) = 0$$

$$(y - 17)(y - 8) = 0$$

$$y = +17 \text{ and } +8$$

The following equation is formed from the roots of the above-mentioned equations I, and II is

$$\text{e). } z^2 - 29z + 204 = 0$$

Let

$$z^2 - 17z - 12z + 204 = 0$$

$$z(z - 17) - 12(z - 17) = 0$$

$$(z - 17)(z - 12) = 0$$

$$z = +17 \text{ and } +12$$

Hence, the correct answer is option E.

17. Questions

Answer: D

$$3x^2 - 56x + L = 0$$

$$3 * (14/3)^2 - 56 * 14/3 + L = 0$$

$$196/3 - 784/3 + L = 0$$

$$L = 588/3$$

$$L = 196$$

Let

$$3x^2 - 56x + 196 = 0$$

$$3x^2 - 42x - 14x + 196 = 0$$

@QuizByMickey

$$3x(x - 14) - 14(x - 14) = 0$$

$$(3x - 14)(x - 14) = 0$$

$$x = +14 \text{ and } +14/3$$

$$\text{So, } M = 14$$

I: The HCF of L and M is 50% more than the value of N.

Let

$$\text{HCF of L and M} = 14 \text{ and } 196 = 14$$

$$150\% \text{ of } 14/3 = 7$$

This is not satisfied.

II: The value of L is a perfect square.

Let

$$L = 196$$

This is satisfied.

$$\text{III: } L = M^2$$

Let

$$196 = 14 * 14$$

$$196 = 196$$

This is satisfied.

Hence, the correct answer is option D.

18. Questions

Answer: A

$$G + 2x^2 - (x - 7)^2 = 37x + 7$$

$$G + 2x^2 - (x^2 + 49 - 14x) = 37x + 7$$

$$G + x^2 - 23x - 56 = 0$$

The smaller root of the equation is 9

$$G + 81 - 23 * 9 - 56 = 0$$

$$G - 182 = 0$$

$$G = 182$$

Let

$$182 + x^2 - 23x - 56 = 0$$

@QuizByMickey

$$x^2 - 23x + 126 = 0$$

$$x^2 - 14x - 9x + 126 = 0$$

$$x(x - 14) - 9(x - 14) = 0$$

$$(x - 9)(x - 14) = 0$$

$$x = +9 \text{ and } +14$$

The larger root of the equation is 14

Thus, the required sum = $14 + 182 = 196$

Hence, the correct answer is option A.

19. Questions

Answer: D

$$\text{II: } 1 - (6/y) - (475/y^2) = 0$$

Let

$$1 - (6/y) - (475/y^2) = 0$$

Multiply by ' y^2 '

$$y^2 - 6y - 475 = 0$$

$$y^2 - 25y + 19y - 475 = 0$$

$$y(y - 25) + 19(y - 25) = 0$$

$$(y - 25)(y + 19) = 0$$

So, $y = -19$ and 25

Then

The positive root of equation II is 25

So

Equation I

$$(220.5/x) + (272.5/y) = 23.15$$

Let

$$(220.5/x) + (272.5/25) = 23.15$$

$$(220.5/x) + 10.9 = 23.15$$

$$220.5/x = 12.25$$

$$x = 220.5/12.25$$

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$$x = 18$$

Hence, the correct answer is option D.

20. Questions

Answer: A

$$\text{I: } x^2 - 7x - 330 = 0$$

Let

$$x^2 - 22y + 15x - 330 = 0$$

$$x(x - 22) + 15(x - 22) = 0$$

$$(x - 22)(x + 15) = 0$$

$$x = 22 \text{ and } -15$$

$$\text{II: } y^2 - my + 462 = 0$$

The highest root in equation I = 22

So

$$22^2 - m * 22 + 462 = 0$$

$$22m = 946$$

$$m = 43$$

So

$$y^2 - 43y + 462 = 0$$

$$y^2 - 22y - 21y + 462 = 0$$

$$y(y - 22) - 21(y - 22) = 0$$

$$(y - 22)(y - 21) = 0$$

$$y = +22 \text{ and } +21$$

So, another root of the equation II is 21

Hence, the correct answer is option A.

@QuizByMickey